Jonathan Giboy Panicker

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EDUCATION

Muthoot Institute of Technology and Sciences — **Expected: June 2029**
BTech Computer Science Engineering — *Kochi,India*

Primus Public School — **June 2023-May 2025**
Karnataka Pre University Board, PCM+CS — *Bengaluru,India.*

Bethany High School — **June 2010-May 2023**
Indian Certificate of Secondary Education — *Bengaluru,India.*

EXPERIENCE

MITS Polaris Game Labs — **June 2026 – Present**
Design Lead — *Remote*

- Designed posters, social media posts, and stories for club events and promotions.
- Assisted in organising and coordinating events, contributing to event planning and execution.

MunWala — **March 2023 – December 2024**
Social Media Designer Intern — *Remote*

- Handled the designing of various stories and posts

PROJECTS

CHIP-8 & Super-CHIP Emulator

- Implemented a CHIP-8/SCHIP virtual machine with instruction decoding and execution, graphics rendering, keyboard input, timers, and audio.
- Developed a live debugging environment with a disassembler, breakpoints, execution tracing, memory inspection, and register-level change tracking.
- Engineered a ROM assembler, versioned binary save-state system, configurable compatibility modes, and persistent user flags.
- Built an interactive SDL2 and Dear ImGui interface with customizable rendering palettes, keyboard layouts, and waveform-based audio synthesis.

LLM Defensive Safety & Reliability Layer

- Designed a modular defensive framework to improve LLM reliability by addressing jailbreaks, prompt injection, hallucinations, and unsafe tool execution.
- Developed a multi-layered input defense pipeline using semantic attack pattern detection, obfuscation analysis, and trust-boundary enforcement.
- Designed agent security controls incorporating input sanitization, tool-call authorization, and provenance tracking to improve the safety and reliability of AI systems.

Credit Default Risk Prediction System

- Built a modular, end-to-end ML pipeline for retail loan default prediction, covering automated preprocessing, model training, and risk classification.

Medemi AI

- Built an offline medical triage application using Python and NLP-based symptom extraction, paired with rule-based clinical logic to classify patient urgency and recommend medical specialties.
- Designed the full application workflow spanning symptom input, urgency assessment, and doctor recommendation/booking.

Navu | Tongue-Controlled Mouse

- Built a real-time HCI system in Python using OpenCV and MediaPipe (468 3D facial landmarks) to map sub-pixel tongue displacements to native Windows cursor coordinates via the User32 CTypes API.
- Implemented HSV/LAB color-space segmentation with an Exponential Moving Average (EMA) filter to eliminate input jitter and enable precision spatial tracking with trigger-based double-clicks.

SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, Bash, Verilog HDL, LEAN
Web / Backend: Flask, Django, React.js, Streamlit, REST APIs,
Database: SQL, MongoDB
AI / ML: Machine Learning, NLP, NEAT (evolutionary algorithms), Computer Vision (OpenCV, MediaPipe)
Tools: Git/GitHub, VS Code
Design Platforms: Figma, Adobe Photoshop, Adobe Illustrator, Gimp

ACHIEVEMENTS

- Selected as a solo finalist at NIT Calicut's flagship hackathon, Tathacks 2026, from 160+ teams and 500+ participants.
- Won a hackathon hosted by StartupYou at Indus International School during Grade 11